MILITARY COLLEGE OF SIGNALS

DEPARTMENT OF COMPUTER SOFTWARE ENGINEERING

Assignment - 1

Applied Physics	
How can we extract precious materials from computing devices?	
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1. Introduction

Modern computing devices contain many different materials, and some of them are more valuable than we might expect. A desktop computer, laptop, smartphone, server or networking device may contain common engineering materials such as iron, aluminium, copper and plastics, but it can also contain smaller quantities of economically important metals such as gold, silver, palladium, cobalt, lithium, tantalum and rare-earth elements. These materials are not randomly distributed; each component uses particular materials because of electrical conductivity, corrosion resistance, magnetic properties, optical behaviour, heat resistance or chemical stability.

When a computer or other electronic device is no longer useful, it usually becomes part of the electronic-waste, or e-waste, stream. E-waste is therefore not simply rubbish. It is a secondary source of raw materials. This is often called urban mining. The basic idea is simple: instead of getting all metals from newly mined ores, we can recover some of them from products that have already been used.

The Global E-waste Monitor 2024 reported that the world generated about 62 million tonnes of e-waste in 2022. Only 22.3% was documented as formally collected and recycled. The same assessment estimated that around 31 million tonnes of metals were embedded in that e-waste, with an estimated combined metal value of about US$91 billion. These figures show why recovery is both an environmental issue and an economic opportunity.

Physics is important here because different materials behave differently. Those differences can be used to separate them. Magnetic separation uses differences in magnetic response; screening and gravity methods use particle size and density; electrical separation uses conductivity and charge behaviour; thermal processing uses differences in melting and chemical stability; and hydrometallurgical processes use differences in solubility and chemical reactivity. Industrial recycling combines several of these principles rather than relying on a single method.

In this assignment, I will look at where useful materials are found in computing devices, how they are recovered in recycling facilities, and why this recovery matters for the economy and the environment. The emphasis is on safe industrial-scale principles rather than household chemical extraction. Some recovery processes involve corrosive chemicals, high temperatures or hazardous emissions and therefore require controlled facilities, trained personnel and appropriate waste treatment.

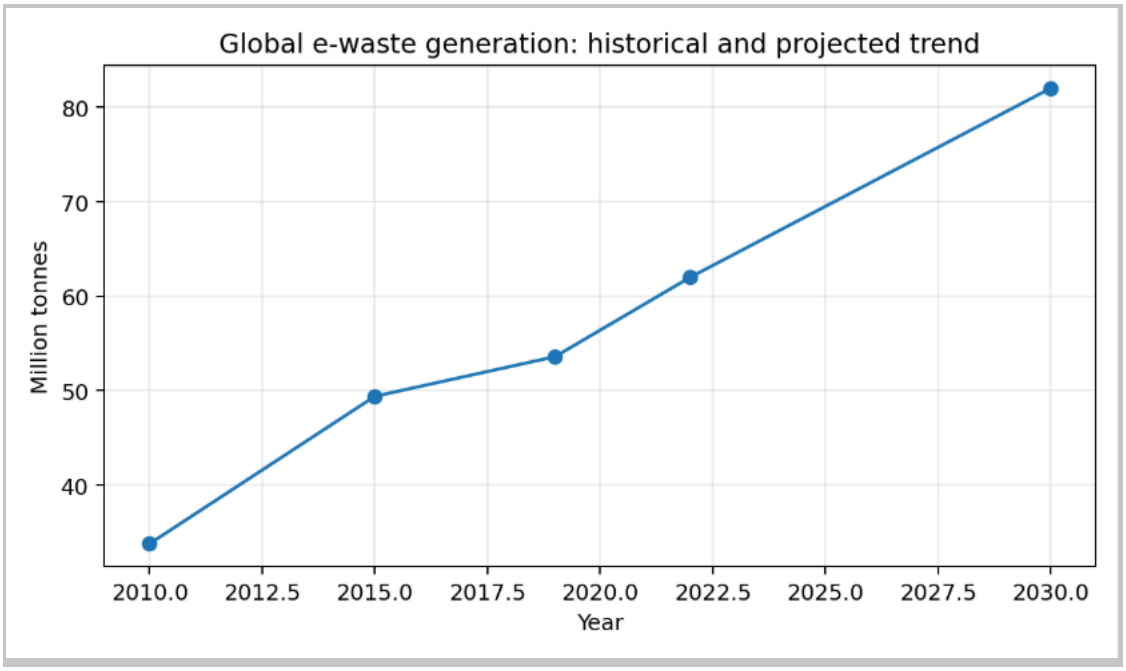


Figure 1. Global e-waste generation has increased substantially, reaching about 62 million tonnes in 2022; the 2030 value is a projection.

2. Precious Materials from different Components

Different parts of a computer are made for different purposes, so their materials are also different. A printed circuit board contains conductive tracks, solder, electronic components and protective layers. A battery is designed around electrochemical reactions and therefore contains electrode and electrolyte materials. A hard disk contains strong permanent magnets, while a display may contain transparent conductive coatings. Understanding the component first makes it easier to understand the recovery process later.

Components	Important Materials	Why they are used/recovered
PCBs	Cu, Au, Ag, Sn, Ni, Pd, Fe, Al, plastics, ceramics	Conductive tracks, contacts, solder and electronic component structures
ICs / chips	Si, Cu, Al, Sn; small Au in some packages	Semiconductor function and electrical interconnection

Connectors	Au, Cu, Ni, Sn	Low-resistance, corrosion-resistant electrical contacts
Capacitors	Ta, Ni, Cu, Ag, Pd in some types	Capacitance, stable electrical behaviour and compact packaging
Batteries	Li, Co, Ni, Mn, Cu, Al, graphite	Electrochemical energy storage
Displays	Glass, In, Sn and other materials	Transparent conductive layers and display structure
Speakers/microphones	Nd, Pr, Dy, Fe, Cu	Permanent magnets and electrical coils
HDDs	Rare-earth magnets, Al, Fe, Cu	High-strength magnets, casing and motor/electrical parts
PSUs/VRMs	Cu, Al, Fe, Sn, Si	Power conversion, heat dissipation and magnetic components
LEDs/semiconductors	Ga, In, Si and rare-earth phosphors	Light emission and semiconductor operation

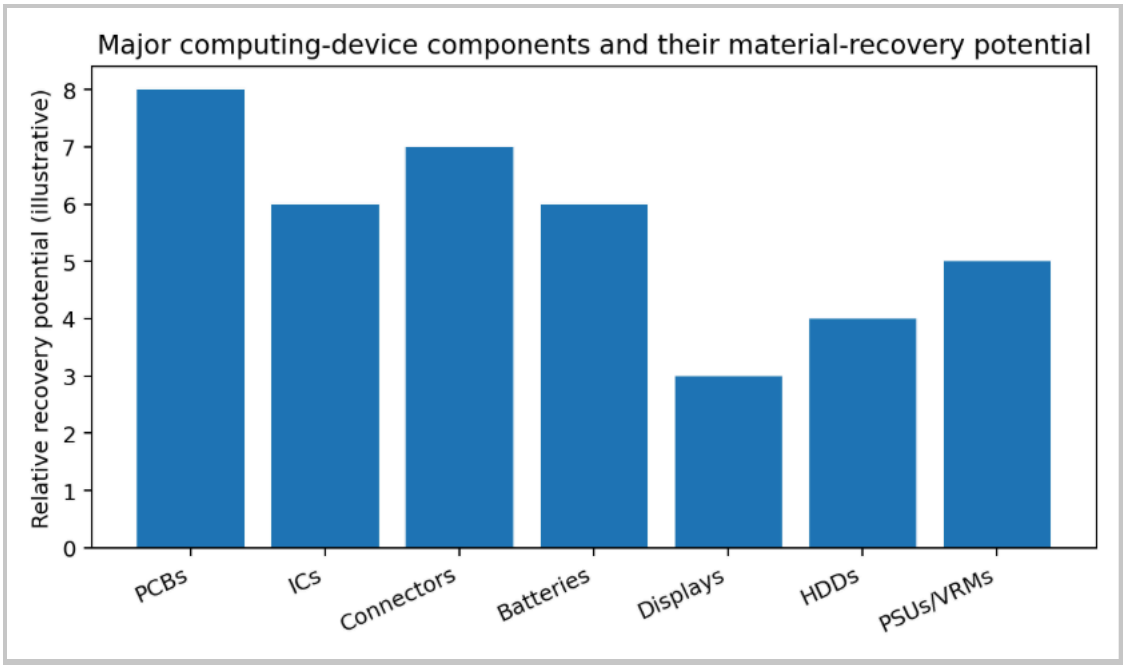


Figure 2. Major computing-device components differ in their potential for material recovery. The ranking shown is illustrative, not a measured metal concentration.

2.1 Printed Circuit Boards (PCBs)

Printed circuit boards are among the most important targets for e-waste recycling because they combine many materials in a compact structure. Copper is used extensively in conductive tracks and layers. Tin is common in solder, while aluminium, iron, nickel and zinc can occur in different components. Gold and other precious metals may occur in very small quantities on contacts, bonding structures or plating. A review in RSC Advances notes that end-of-life PCBs can contain many different elements and may have a relatively high metal fraction compared with ordinary mined ores.

Industrial recycling normally begins with removal of large components and mechanical size reduction. The resulting material can then be separated according to physical properties before metal-rich fractions are processed by thermal, chemical or electrochemical routes. The important point is that valuable metals cannot simply be removed in one step. They first have to be separated and concentrated, and then purified. Plastic and ceramic fractions must also be managed because they affect the thermal and chemical processes.

2.2 ICs / Chips

Integrated circuits contain semiconductor material, usually silicon, together with metallic interconnects and packaging materials. Copper and aluminium are important conductors, while solder and package materials can contain tin and other metals. Some older or specialised packages use gold in bonding wires or plated connections. Because a single chip contains very small amounts of valuable metals, economic recovery generally depends on collecting large quantities rather than processing individual chips.

The physical structure of semiconductor packages also makes separation challenging. A recycling facility may first remove chips from boards, classify the material and send it into a metal-concentration process. The recovered fractions can then be refined further. This is one reason why recycling is done on a large scale. A tiny amount from one chip may not be worth much, but the same amount repeated across tonnes of material can become valuable.

2.3 Connectors and Contacts

Connectors are designed to provide reliable electrical contact over many connection cycles. Copper or copper alloys provide the conductive base, while nickel can act as a barrier layer and tin or gold may be used on contact surfaces. Gold is especially useful because it is highly resistant to corrosion and maintains good electrical contact at very small contact forces.

In recycling, connectors can be separated from boards and processed as a relatively metal-rich fraction. Mechanical sorting is important because unnecessary mixing with plastics lowers the

concentration of recoverable metals. Industrial refiners then use controlled chemical or metallurgical processing to separate base metals from precious-metal-bearing fractions.

2.4 Capacitors

Capacitors store electrical energy and are found throughout computers, power supplies and motherboards. Different capacitor technologies use different materials. Tantalum capacitors are particularly important because tantalum is a critical material with specialised electrical properties. Other capacitors can contain nickel, copper, aluminium, silver or, in some designs, palladium-bearing materials.

The recycling route depends strongly on capacitor type. Components are first identified and separated where economically justified. The material may then enter a controlled metallurgical process in which metals are concentrated and refined. Because capacitor compositions vary, modern recycling systems benefit from good sorting and product knowledge.

2.5 Batteries

Rechargeable batteries are an important part of modern portable computing devices. Lithium-ion batteries may contain lithium, cobalt, nickel, manganese, copper and aluminium, together with graphite and organic electrolyte materials. The exact composition depends on the battery chemistry. Cobalt and nickel can have significant economic value, while lithium is strategically important because of growing demand for rechargeable energy storage.

Battery recycling is different from ordinary PCB recycling because the battery is an electrochemical device and can retain stored energy. Industrial systems therefore begin with safe collection, identification, discharge or other controlled preparation, followed by dismantling and mechanical processing. The active material can form a powder commonly called black mass. Hydrometallurgical or other controlled processes can then recover selected metals. Battery recycling must also address fire risk, electrolyte residues and safe handling.

2.6 Displays / Screens

Displays contain several layers, including glass, transparent conductive materials, electronics and, depending on the technology, liquid-crystal or light-emitting structures. Indium is important because indium-tin oxide is widely used as a transparent conducting material in many display technologies. The quantity in an individual screen is small, but the material has strategic importance because primary indium resources are limited and are strongly associated with zinc processing.

Screen recycling is difficult because the valuable coating is extremely thin and is attached to a large mass of glass and other materials. Effective recovery therefore depends on separating the screen into appropriate fractions and using processes capable of concentrating the desired material. Safe dismantling is also necessary because displays may contain fragile components and, in older technologies, substances requiring controlled handling.

2.7 Hard Disk Drives (HDD)

Hard disk drives contain aluminium and steel structures, copper conductors and powerful permanent magnets. The magnets are especially interesting because some contain rare-earth elements. HDDs also have precisely engineered platters, motors and actuator mechanisms, making them a concentrated source of several engineering materials.

Recycling usually starts with secure collection and mechanical dismantling. Magnetic components can be separated from the rest of the drive, while aluminium, steel and copper can enter conventional metal-recycling streams. Recovery of rare-earth elements from magnets is more specialised and requires dedicated processing if the magnets are not reused directly.

2.8 Power Supplies and VRMs

Power supply units and voltage-regulation modules convert electrical energy into voltages required by computing circuits. They contain copper windings, aluminium heat sinks, steel structures, silicon-based semiconductor devices, solder and magnetic materials. Their relatively high copper content makes them useful to conventional metal recycling as well as specialised e-waste recovery.

Power electronics also demonstrate the importance of design for recycling. Components are physically interconnected, and some are attached using solder or adhesives. Industrial dismantling separates the major fractions before shredding and concentration. Copper-rich fractions can then be refined, while ferrous and aluminium materials can be recovered using established metal-recycling routes.

2.9 LEDs and Other Semiconductor Devices

LEDs and specialised semiconductor devices can use materials such as gallium, indium, silicon and rare-earth phosphors. These materials enable particular optical or electronic properties that ordinary structural metals cannot provide. Their concentrations are generally small, but they can be important because some are critical or strategically significant materials.

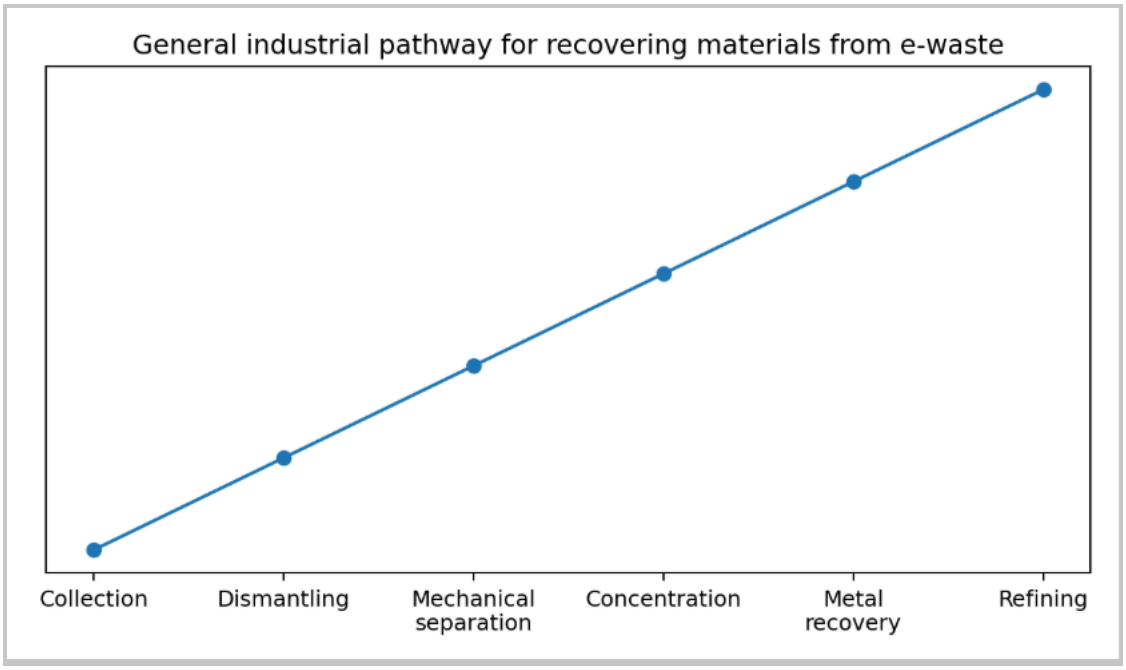
Recovery is technically challenging because the useful material may exist in thin semiconductor layers or in small quantities within complex packages. Consequently, specialised recycling is

more likely to be economically attractive for large, concentrated waste streams. The general strategy is to separate devices, concentrate the relevant fraction and apply a refining process appropriate to the target material.

3. Extraction of Various Materials

Material extraction from e-waste is best understood as a chain of separation and purification steps. A recycling plant does not normally take a mixed computer and immediately obtain pure gold or copper. Instead, the feed is collected, sorted, dismantled, size-reduced and separated into fractions. Each fraction is then sent to the most appropriate recovery route.

3.0 General Industrial Extraction Pathway



Stage	Main Purpose	Typical Physical/Engineering Principle
Collection & sorting	Gather compatible waste streams	Classification and logistics
Dismantling	Remove batteries, drives, boards and other major parts	Mechanical separation

Size reduction	Liberate materials from composite structures	Cutting, crushing, shredding
Mechanical concentration	Separate broad material classes	Magnetic, screening, density and electrical methods
Metallurgical recovery	Transfer target metals into recoverable phases	Pyrometallurgy or hydrometallurgy
Electrochemical recovery	Produce high-purity metals where suitable	Electrowinning/electrorefining
Residue treatment	Handle non-metallic and hazardous fractions	Controlled waste and emissions management

The three major metallurgical approaches are pyrometallurgy, hydrometallurgy and electrometallurgy. Pyrometallurgy uses controlled high-temperature reactions to melt or transform materials and concentrate selected metals. Hydrometallurgy uses aqueous chemistry to dissolve selected materials and then separate them from the solution. Electrometallurgy uses electrical energy to deposit or refine metals. Industrial plants may combine all three.

The following subsections describe the principal materials relevant to computing devices. The descriptions intentionally focus on the physical and industrial principles. Household attempts to recover metals using acids, burning or other uncontrolled processes are unsafe and are not an appropriate recycling method.

3.1 Copper (Cu)

Copper is one of the most abundant valuable metals in electronics. It is used for PCB tracks, cables, connectors, transformer windings, motors and heat-management structures. Its high electrical conductivity makes it indispensable in computing hardware.

Copper-rich fractions can be concentrated mechanically and then treated by industrial smelting or leaching followed by electrochemical purification. In electrowinning or electrorefining, electrical current drives copper ions toward an electrode, producing refined copper. The large amount of copper in e-waste makes it one of the most economically attractive recovery targets.

3.2 Gold (Au)

Gold is used in small amounts where excellent corrosion resistance and reliable electrical contact are required. It can occur on connector surfaces, contact pads and some bonding structures.

Although the total mass is small compared with copper, its high economic value makes it important.

Industrial gold recovery normally begins by concentrating precious-metal-bearing fractions. Controlled chemical or metallurgical processing then separates gold from base metals and other precious metals. The final refining stage can produce high-purity gold. RSC Advances literature reports that e-waste can contain gold concentrations far above those found in many conventional ores, which explains the interest in urban mining.

3.3 Silver (Ag)

Silver has excellent electrical and thermal conductivity and is used in contacts, conductive materials and some electronic components. It can occur together with copper and other metals in circuit assemblies.

After concentration, silver can be recovered from appropriate metallurgical streams. Depending on the feed, it may be separated from copper and other metals during refining. Because silver often occurs alongside other valuable metals, integrated processing is more efficient than treating it as an isolated material.

3.4 Palladium (Pd)

Palladium is a platinum-group metal valued for corrosion resistance and catalytic properties. Electronics have used palladium in certain contacts, multilayer ceramic capacitor systems and specialised components.

Palladium recovery generally occurs from concentrated precious-metal-bearing fractions. Industrial refiners use selective chemical and metallurgical separation to distinguish it from gold, silver, copper and other platinum-group metals. Its high value means that even small quantities can contribute significantly to recycling economics.

3.5 Platinum (Pt)

Platinum is not usually present in large quantities in ordinary computing equipment, but it may occur in specialised electronics and associated industrial components. It is a valuable platinum-group metal with high chemical stability.

Where platinum-bearing material is present in a sufficiently concentrated stream, specialist refiners can separate it from other precious metals. The economic case depends heavily on feed composition and scale.

3.6 Tin (Sn)

Tin is widely associated with solder, which joins components to printed circuit boards. Modern electronics use various lead-free solder formulations, commonly involving tin with other metals.

Tin-bearing fractions can be separated during metallurgical processing. Recovery must account for solder composition and contamination by other metals. The importance of tin also shows that recycling is not limited to expensive precious metals; common metals can represent large total material flows.

3.7 Nickel (Ni)

Nickel is used as a coating, structural material and battery electrode material. It provides corrosion resistance and useful mechanical and electrochemical properties.

Nickel may be recovered from appropriate metal concentrates or battery-derived streams. In battery recycling, nickel recovery is closely linked with cobalt, manganese and other electrode materials, so separation is generally part of a multi-metal process.

3.8 Cobalt (Co)

Cobalt has been important in several lithium-ion battery chemistries because of its role in cathode materials. It also has specialised uses in alloys and electronics.

Industrial battery recycling can recover cobalt from processed battery material. Hydrometallurgical routes are often studied because different metals can be selectively separated from solution. The economic importance of cobalt also provides a strong reason to improve battery collection and recycling.

3.9 Lithium (Li)

Lithium is a key material in rechargeable batteries. Its electrochemical properties allow lithium-ion cells to store substantial energy relative to their mass.

Lithium recovery from spent batteries requires controlled processing because lithium is distributed through electrode materials and other cell components. Modern recycling research includes hydrometallurgical and direct-recycling approaches intended to recover lithium while preserving as much material value as possible.

3.10 Rare-Earth Metals

Rare-earth elements such as neodymium, praseodymium and dysprosium can occur in strong permanent magnets used in drives, speakers and other equipment. They are important because of their magnetic properties and strategic supply considerations.

Recovery is challenging because the elements occur in relatively small quantities and are dispersed among many products. Magnet collection and concentration are therefore essential. Specialist processes can then separate rare-earth elements, while reuse of intact magnets may sometimes be more energy-efficient than breaking them down completely.

4. Economic Effect of E-Waste

E-waste has economic value mainly because there is so much of it and because it contains many different materials. A single old computer may not contain enough recoverable material to justify a complex industrial process by itself. Millions of devices, however, create a large secondary resource stream. So the real question is not just how much gold or copper is inside one device. The bigger question is whether enough devices can be collected and processed at a reasonable cost.

4.1 Size of E-Waste

The Global E-waste Monitor 2024 estimated global e-waste generation at 62 million tonnes in 2022. It also projected that annual generation could reach about 82 million tonnes by 2030 if current trends continue. This growth is associated with increasing consumption of electrical and electronic equipment, shorter replacement cycles for some products, technological change and limited repair or reuse in some markets.

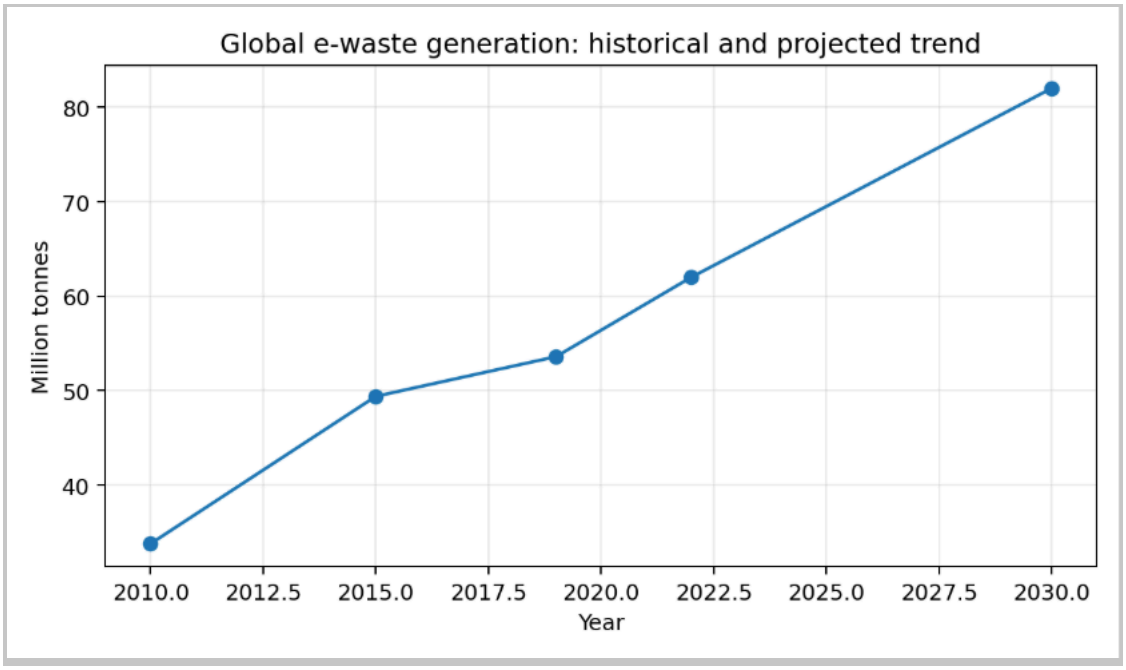


Figure 4. Global e-waste generation reached about 62 Mt in 2022; 82 Mt is projected for 2030 in the cited WHO/Global E-waste Monitor context.

4.2 Metals Inside It

Of the 62 million tonnes generated in 2022, the Global E-waste Monitor estimated about 31 million tonnes of metals embedded in the waste stream. This illustrates a major difference between e-waste and ordinary municipal waste: a substantial portion of the discarded mass is made of materials that already have established industrial value.

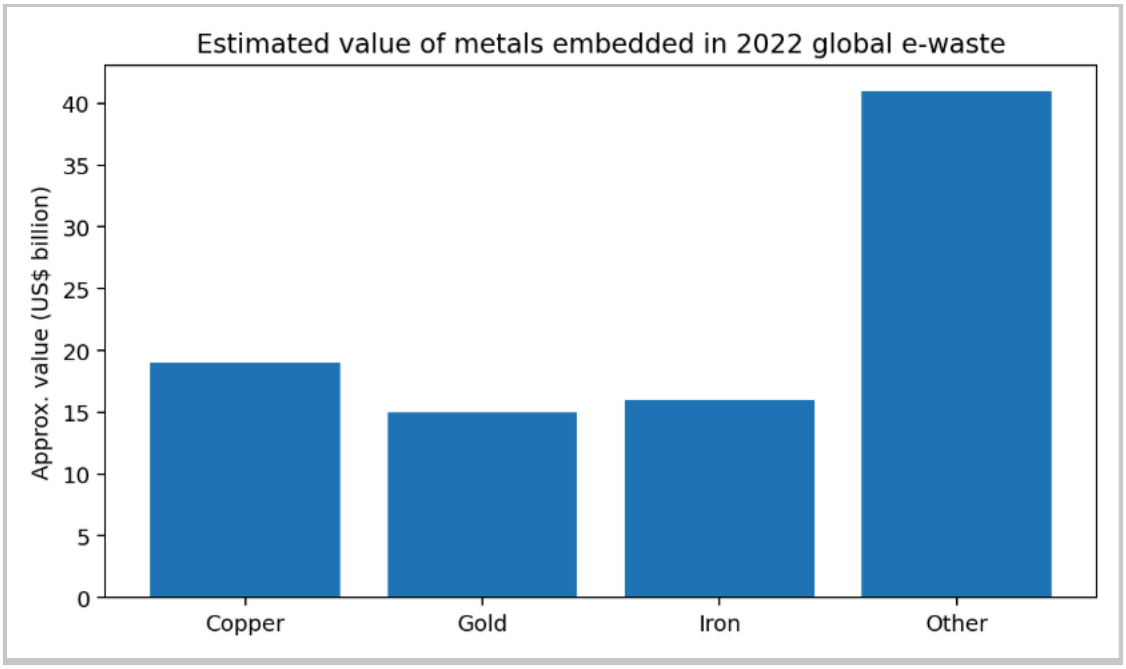


Figure 5. Selected estimated values of metals embedded in 2022 global e-waste. The figures are approximate values reported in the Global E-waste Monitor context.

Indicator (2022)	Reported estimate	Meaning
Total e-waste	62 Mt	Mass of electronic and electrical waste generated globally
Metals embedded	31 Mt	Large secondary source of metallic materials
Estimated metal value	US$91 billion	Approximate economic value of embedded metals
Copper value	US$19 billion	One of the major value contributors
Gold value	US$15 billion	High value despite relatively small mass
Iron value	US$16 billion	Large contribution from common structural metal
Secondary raw materials reclaimed	~US$28 billion	Value associated with documented recycling/urban mining

4.3 Recovering Value

Recovering value from e-waste requires more than knowing that a metal is present. Collection, transportation, labour, energy, equipment, refining, environmental controls and residue treatment all have costs. The economic value of recycling is therefore determined by the difference between the value of recovered materials and the total cost of obtaining them safely.

High-value metals such as gold and palladium can improve the economics of processing circuit boards, while large quantities of copper, aluminium and iron provide additional revenue. In some cases, reusable components can create value before the material even reaches metal refining. This is why modern circular-economy systems consider repair, refurbishment, component reuse and material recycling as connected stages rather than completely separate activities.

4.4 Industry

The e-waste recycling industry contains several levels. Collection companies gather discarded products. Dismantlers remove batteries, circuit boards, drives and other components. Mechanical processors shred and separate material streams. Specialist metallurgical companies refine concentrated fractions. Manufacturers can then purchase the recovered secondary raw materials.

Examples of industrial practice demonstrate that specialised facilities can recover several metals from electronic scrap. C-MET has described a demonstration-scale PCB processing facility using steps including depopulation, shredding, smelting, remelting, electrorefining, leaching and gas cleaning. Industrial companies such as Umicore also process e-scrap and recover precious and base metals. These examples show that recovery is a coordinated industrial operation rather than a single extraction reaction.

4.5 Effect on Employment

Recycling creates employment in collection, transportation, sorting, dismantling, equipment operation, laboratory analysis, maintenance, environmental monitoring and administration. More advanced facilities also require engineers, chemists, metallurgists and technicians. The industry can therefore support both manual and highly technical jobs.

Employment estimates must be interpreted carefully. For example, broader U.S. EPA figures for recycling and reuse have been used to illustrate the scale of employment associated with the wider recycling economy, but those figures are not an e-waste-only employment count. A strong e-waste sector can nevertheless contribute to local employment when collection and processing infrastructure is developed.

4.6 Recycling vs Mining

Primary mining extracts material from geological deposits, while urban mining recovers material already incorporated into products. Recycling can reduce the need for some new extraction,

especially when recovered metals meet industrial quality requirements. The advantage is strongest for materials that are energy-intensive to mine and refine or that occur in relatively concentrated electronic fractions.

The Global E-waste Monitor estimated that documented e-waste recycling in 2022 avoided the need for approximately 900 million tonnes of primary ore extraction. This does not mean that recycling completely replaces mining. New materials are still needed, and recycling itself requires energy and infrastructure. Instead, recycling supplements primary resources and can reduce pressure on mines and ecosystems.

4.7 Factor of Scale

The size of the waste stream matters a lot in e-waste recycling. A single mobile phone may contain only fractions of a gram of a valuable metal, but millions of phones create a feedstock measured in tonnes. Large quantities allow expensive sorting, analytical equipment and refining plants to be used efficiently.

Scale also improves logistics. A recycling plant can operate continuously when it receives predictable quantities of compatible material. This reduces the cost per kilogram and makes it easier to justify investment in pollution-control systems and worker-safety measures.

4.8 Wider Economic Picture

The wider economic benefit of e-waste recycling includes reduced dependence on imported raw materials, development of recycling technologies, creation of skilled employment, recovery of materials for manufacturing and reduction of environmental cleanup costs. Countries that develop reliable collection and processing systems can retain more material value inside their economies.

The World Health Organization has also highlighted the economic significance of better e-waste management. Its discussion of reaching a 60% collection and recycling rate by 2030 indicates that the global benefits could exceed costs by more than US$38 billion. Such estimates reinforce the idea that environmental management and economic development can support each other when recycling systems are properly designed.

4.9 Informal Recycling

Informal recycling is an important part of the global e-waste landscape, especially where formal collection systems are weak. Informal workers may recover valuable components from discarded electronics, but unsafe methods can expose workers and nearby communities to hazardous

substances. The World Health Organization identifies activities such as open burning, uncontrolled heating, acid treatment, dumping and unsafe dismantling as significant concerns.

The solution is not simply to eliminate livelihoods without alternatives. A safer approach is to connect collection networks with formal recycling facilities, provide training and protective measures, improve traceability and create economic pathways into regulated processing. Formalisation can preserve the useful collection function of informal workers while reducing dangerous exposure.

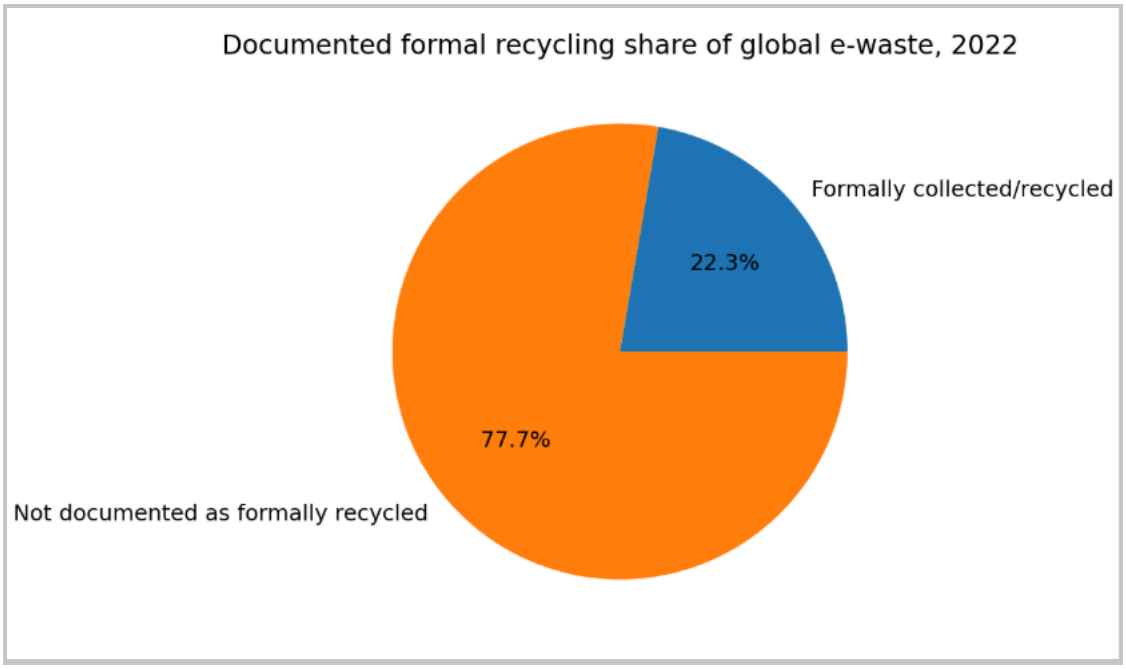


Figure 6. In 2022, only 22.3% of global e-waste was documented as formally collected and recycled.

5. Conclusion

Computing devices are complex assemblies of materials that remain valuable even after the devices themselves stop working. Printed circuit boards contain copper and smaller quantities of precious metals; batteries contain strategically important electrochemical materials; hard disk drives and speakers can contain rare-earth magnets; displays can contain specialised transparent conductors; and power electronics contain large amounts of copper and aluminium. Together, these components make e-waste an important secondary resource.

The recovery of these materials is based on fundamental engineering and physical principles. Mechanical methods separate components and particle sizes. Magnetic and electrical methods exploit differences in material properties. Thermal processing concentrates and transforms

metals, while hydrometallurgical and electrochemical methods allow selected metals to be separated and purified. Industrial recycling therefore represents an application of physics, chemistry, materials science and engineering in a single process chain.

The scale of the problem makes recycling increasingly important. About 62 million tonnes of e-waste were generated globally in 2022, while only 22.3% was documented as formally collected and recycled. Around 31 million tonnes of metals were embedded in this waste, with an estimated value of about US$91 billion. Recovering even a larger fraction of these materials can reduce pressure on primary resources, create economic activity and support a more circular materials economy.

At the same time, e-waste recycling must be carried out safely. Uncontrolled burning, dumping or chemical treatment can create serious health and environmental hazards. The appropriate approach is therefore formal collection, professional dismantling, controlled industrial recovery and proper treatment of residues and emissions. In this way, discarded electronics can be transformed from an environmental burden into a useful source of secondary raw materials.

6. References

UNITAR & ITU. Global E-waste Monitor 2024: Electronic Waste Rising Five Times Faster than Documented E-waste Recycling. 2024.

2. World Health Organization (WHO). Electronic waste (e-waste). Fact sheet, 1 October 2024.
3. World Health Organization (WHO). E-waste and child health: information on hazards and health effects of unsafe recycling.
4. World Health Organization (WHO). Electronic waste: Digital dumpsites and children's health.
5. Rao, M. D., Singh, K. K., Morrison, C. A., & Love, J. B. (2020). Challenges and opportunities in the recovery of gold from electronic waste. RSC Advances, 10, 4300–4309. DOI: 10.1039/C9RA07607G.
6. C-MET. Demonstration-scale electronic waste/PCB processing information, including depopulation, shredding, smelting, remelting, electrorefining, leaching and gas cleaning.
7. Umicore. E-scrap recycling and recovery of precious and base metals from electronic scrap.
8. Royal Mint. Information on recovery of precious metals and other materials from electronic scrap.
9. U.S. Environmental Protection Agency (EPA). Recycling and reuse economic impact information; broader recycling/reuse employment figures cited with appropriate scope caveat.